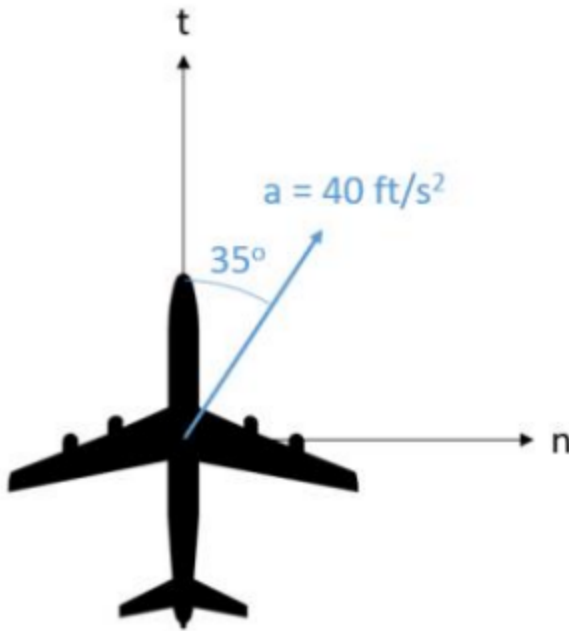


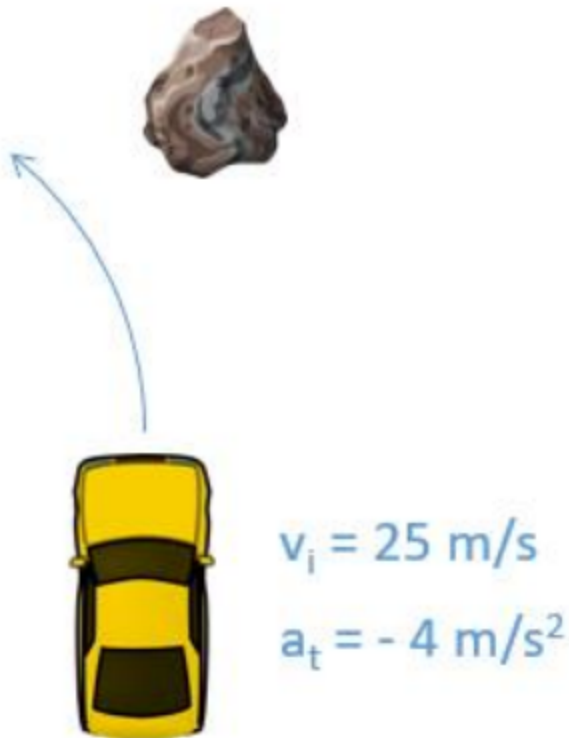
Lesson 4: 2D motion with rectangular coordinates

Lesson 4: 2D motion with Normal-Tangential coordinates

1. A plane with a current speed of 600 ft/s is increasing in speed while also making a turn. The acceleration is measured at 40 ft/s^2 at an angle 35° from its current heading. Based on this information, determine the rate at which the plane is increasing its speed (tangential acceleration) and the radius of the turn for the plane. /1



2. A car traveling 15 m/s notices a fallen rock in the road ahead. It /1
immediately begins decreasing its speed at a rate of 4 m/s^2 and at
the same time begins turning to the left as shown below. If the tires
will allow for an overall acceleration of no more than 6 m/s^2 , what
is the minimum radius for the turn before the car will lose grip?



Answer: _____

3. A roller-coaster car is traveling at a speed of 20 m/s when it passes through point B . At that point, it enters a concave down circular section of the track that has a radius of curvature of 60 m . After it reaches point B , the car's speed increases at a rate of 4 m/s^2 . Determine (a) the time it takes the car to reach point D , (b) the x and y components of the overall acceleration of the car when it reaches point D . /1

